

Primer

Muscular hydrostats

Noel Naughton

In *Harry Potter and the Chamber of Secrets*, Professor Lockhart's inept attempt to heal Harry's broken arm with a spell instead removes all the bones in Harry's arm, leaving a useless floppy arm in its place. While a night of the potion Skele-Gro eventually regrows Harry's bones, this episode neatly highlights the critical role a rigid skeletal support structure plays in our ability to move. Without one, we are stuck. Yet some of the most dexterous and impressive examples of movement in the animal kingdom do so without such rigid support structure. Tongues, tentacles, and trunks, as memorably framed in a seminal paper by William Kier and Kathleen Smith, are in a unique class of appendages known as muscular hydrostats that include octopus arms, elephant trunks, squid tentacles, and many animal tongues (Figure 1). These appendages lack bones or other rigid support entirely and instead consist only of soft tissue such as muscle and connective tissue.

To understand how these animals can generate movement without bones, it helps to first step back and ask a broader question: how do animals generate movement at all? As Newton taught us with his three laws,

generating motion requires forces. At the organismal scale, the forces used for nearly all animal motion are produced by muscle fibers (though exceptions abound, especially in smaller organisms). But muscle produces a linear, one-dimensional contractile force. To translate this into a useful three-dimensional motion, the force must be resisted and redirected. A vertebrate's bony endoskeleton or an arthropod's chitinous exoskeleton helps structure and transmits the one-dimensional contractile forces produced by soft muscle tissue to generate useful movement. This combination of hard and soft materials also solves a fundamental problem: muscle can only pull, it cannot push. Rigid bones allow antagonistic muscles, such as your arm's biceps and triceps, to move a joint in both directions. Indeed, as Harry Potter found out, without a humerus to provide structure, we would each get to bend our elbow only once before it would be curled up for good.

Moving without a rigid skeleton

So how do muscular hydrostats do it? To understand, let us first consider a class of closely related structures known as hydrostats or hydrostatic skeletons that have also evolved to generate motion without a permanent rigid skeleton. Put simply, hydrostats consist of a fluid-filled cavity surrounded by a fibrous tissue membrane that holds the fluid, generally water, in as a bag. Consider squeezing a balloon filled with water and a balloon

filled with air from every direction. The air-filled balloon will compress, shrinking its volume roughly proportional to how hard you squeeze. In contrast, the water-filled balloon is incompressible: it will remain the same volume even as you squeeze harder (in the extreme, water compressed under the intense pressures found at the bottom of the Mariana Trench will shrink ~4–5% while air under that same pressure would shrink >99%). This conservation of fluid volume is the critical property of water that enables hydrostatic skeletons to function. If muscle contractions compress the fluid chamber in one direction, the fluid chamber will extend in another direction to maintain a constant volume, often guided in its response by the orientation of the fibrous connective sheath that surrounds the fluid chamber. For example, if you squeeze a cylindrical water balloon radially, it will extend axially. Additionally, if you squeeze from all sides, the fluid chamber will have nowhere to deform and will instead stiffen its structure to provide dynamic skeletal support — hence their name: 'hydro' + 'static.' Earthworms, sea anemones, and the tube feet of starfish are a few classic examples of this class of hydrostatic soft structure that can dynamically stiffen itself to support movement.

Distinct from, though related to, these classical hydrostats are muscular hydrostats, which forgo a fluid-filled cavity and instead consist entirely of soft tissues. Fundamentally, both



Figure 1. Muscular hydrostats found in nature.

Muscular hydrostats are common in nature; examples include octopus arms, squid tentacles, elephant trunks, and the tongues of giraffes, nectar bats, snakes, and lizards. Muscular hydrostats have independently evolved multiple times, and all operate according to similar biomechanical principles that allow them to forgo rigid bony skeletons while remaining capable of strong and dexterous movement. Octopus adapted from Tekinalp *et al.* (2024). Squid reused from Meressa Chartrand/Unsplash. Elephant reused from Sneha Cecil/Unsplash. Giraffe reused from Slawek K/Unsplash. Snake reused from Badal Kumar/Unsplash. Lizard reused from Andrey Tikhonovskiy/Unsplash. Bat © Nathan Muchhala.

classical hydrostats and muscular hydrostats exploit similar physical principles, so much so that muscular hydrostats explicitly derive their name from hydrostats despite the lack of a fluid-filled chamber. When working with compliant materials that have a constant volume, deforming the material in one direction causes the material to expand or shrink in another direction to conserve its volume. More technically, this requires materials with high bulk moduli and low shear moduli. Hydrostats use incompressible water for this, while muscular hydrostats use soft tissue, which itself is largely made up of water and so is also essentially incompressible. Muscular hydrostats are often described as hydrostats but with muscle instead of water. However, this switch from a fluid to a solid tissue entails key changes, meaning that muscular hydrostat mechanics diverge from those of classical hydrostats.

The main distinction between the biomechanics of hydrostats and muscular hydrostats relates to the fundamental mechanical difference between a fluid and a solid. A fluid is unable to resist shear stress — the stress (force) that develops when it slides or rubs against something — while a solid material can. If you drag your hand along the top of a pool of water, it will begin to flow with your hand. If the pool is filled with honey, your hand will move more slowly due to the higher viscosity, but it will still flow. In contrast, drag your hand along a soft rubber block glued to a table: the rubber will initially deform (shear) with your hand before reaching a limit where the internal elastic forces of the solid resist further deformation.

Because fluid is unable to resist shear, any forces applied equilibrate within the fluid volume as a uniform pressure. The impact and limitations of this can be seen in the hydrostatic skeletons of worms. Some phyla, such as nematodes, have a single body cavity that runs the whole length of the animal. Here, muscle contractions increase the internal pressure of the entire body cavity, limiting localized control. Other worms such as annelids, for example earthworms, solve this constraint by compartmentalizing their body cavity with walls of tissue called septa that localize pressure and so enable local control of deformation. However, such

compartments are anatomically fixed, limiting reconfigurability. In contrast, muscular hydrostats are made up completely of soft tissue, which is able to sustain the internal shear stresses necessary to produce the highly localized, reconfigurable, and complex deformations that octopus arms, elephant trunks, and human tongues are known for.

How do octopus arms work?

But how does a muscular hydrostat actually generate movement? To do so, it must translate the one-dimensional contraction of its muscle fibers into three-dimensional deformations, which it accomplishes through the careful architectural organization of its muscle fibers. This can be best understood by considering the anatomical specifics of one of the best-known muscular hydrostats: the octopus arm.

The arm of an octopus — a fun piece of trivia is that octopuses do not have tentacles; in cephalopods, arms have suckers along the entire stalk while tentacles have suckers on only the club at the end of the stalk — is an excellent example of how the three-dimensional architectural organization of an octopus arm enables its dexterous function. A cross-section of an octopus arm (*Octopus rubescens*) is shown in [Figure 2A](#). Surrounding the neural tissue that runs the length of the arm are three major muscle groups: the longitudinal, transverse, and oblique muscles. The longitudinal muscles are the easiest to understand. They run along the length of the arm and are roughly organized into four groups at the top (aboral), bottom (oral), left, and right side of the arm. If all of the longitudinal muscle groups contract, the arm shortens. If, instead, only some contract, such as those located orally (at the bottom), the muscles' location offset from the central axis of the arm will produce a local torque that generates bending in the arm.

Once contracted, however, these longitudinal muscles, like all muscles, are unable to elongate themselves. Here the arm's lack of a rigid skeleton precludes the traditional antagonist muscle setup (think biceps–triceps again). Instead, transverse muscles lie perpendicular to the longitudinal muscles in the cross-section; located inside of the longitudinal muscles, they extend through them via

thin projections known as trabeculae to connect to the outside of the arm. When they contract they, along with a thin layer of peripheral circular muscle, squeeze the arm radially, causing the arm to extend axially to conserve volume. With these longitudinal and transverse muscle groups, the arm can extend, shorten, and bend. Finally, three layers of oblique muscles on the left and right sides of the arm run helically along the arm. Adjacent layers are wrapped in opposite directions (handedness) from each other. Selective activation of oblique muscles with a specific handedness will then twist the arm, completing the arm's kinematic repertoire.

With these muscle groups, an octopus can bend, elongate, shorten, and twist each of its eight arms locally along its length, which has been confirmed through observations in both the lab and the wild. It is important to note, however, that the octopus likely uses a combination of muscle groups to perform even simple actions such as bending. Contracting multiple muscles allows the arm to generate dynamic stiffening that can amplify a movement or resist undesired deformations. Additionally, experiments have shown the contractile properties of skeletal muscle are influenced by intramuscular pressure, an effect likely also present in the octopus arm, making its biomechanics a complex interplay of shear and pressure effects. Overall, the basic mechanical organization of the octopus arm as a fiber-reinforced solid that lacks a fluid compartment or enclosing membrane to act like a pressure vessel is well understood. What continues to be worked out are the finer details of the material properties of the arm's different muscle groups and connective tissue.

Trunks, tentacles, and tongues

Despite being separated evolutionarily by over half a billion years, there is a striking similarity between the muscular organization of elephant trunks and octopus arms, and indeed between all muscular hydrostats broadly. Like the octopus arm, an elephant trunk consists of longitudinal, transverse, and oblique muscle groups, as well as a radial group that acts similarly to transverse muscles ([Figure 2B](#)). This similarity perhaps reflects the limited ways a one-dimensional muscle

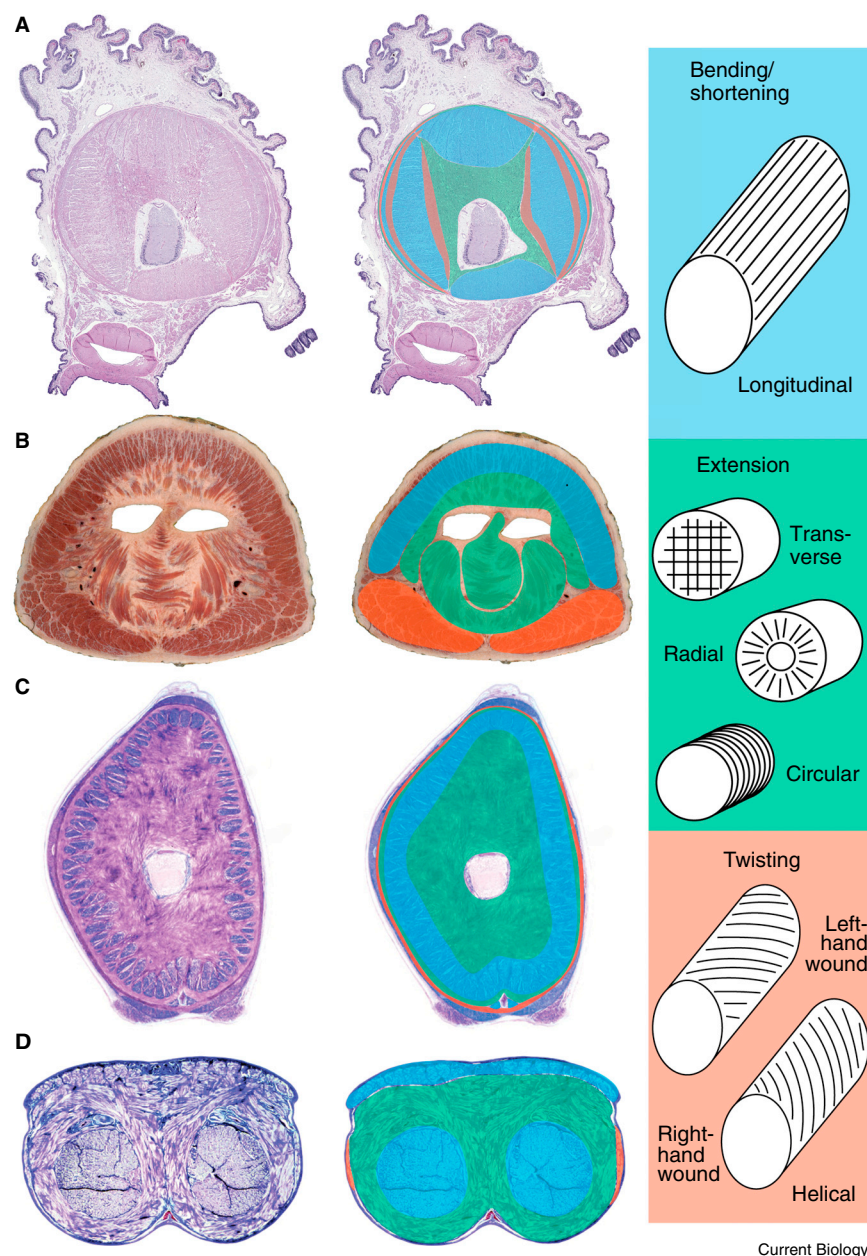


Figure 2. The anatomy of a muscular hydrostat.

Muscular hydrostats are generally made up of several common muscle groups that run longitudinally along their length, are arranged in the cross-section with a transverse, circular, or radial organization, or lie obliquely along a helical path. These core types of organization are found in: (A) octopus arms (*Octopus rubescens*); (B) elephant trunks (*Elephas maximus*), image based on Dagenais *et al.* (2021); (C) squid tentacles (*Doryteuthis pealeii*); and (D) lizard tongues (*Varanus exanthematicus*). (C, D) Adapted from Kier (2012) with permission from *The Journal of Experimental Biology*. The middle column shows the major muscle groupings of each muscular hydrostat colored by their general orientation.

contraction can be translated into the three-dimensional deformations of bend, twist, extension, and shortening that a muscular hydrostat can perform. Morphological differences that do exist appear explainable as adaptations to the unique environmental constraints

these creatures inhabit. Elephants live on land, where gravity constantly pulls their soft trunks down. Octopuses dwell underwater in a nearly weightless environment due to buoyancy effects. The symmetry of an octopus arm's musculature facilitates deformations in

any direction, because each direction is equally available as a result of the low gravity. In contrast, the elephant trunk's dorsoventral asymmetry appears organized to counteract gravity's constant pull.

Similar architectural features are found in many other muscular hydrostats, but often simplified to reflect more specialized functions than the generalist manipulation of octopus arms and elephant trunks. One common function of such specialized muscular hydrostats is rapid extension, seen in tentacles of squid such as *Doryteuthis pealeii*. During hunting, their stalks elongate 50–80% in only 20–40 ms. In these cases, careful control of bending (governed by longitudinal muscles) is less important than the generation of strong extension forces (governed by transverse and circular muscles). As a result, the cross-section of the squid tentacle, while similar to the octopus in the spatial organization of its muscle groups — reflecting its related evolutionary history — has a substantially larger transverse muscle and much smaller longitudinal muscles (Figure 2C). The squid tentacle illustrates another advantage of both muscular hydrostats and hydrostats that arises from their typical morphology of elongate cylinders. This is their ability to exploit the length–width relation of a constant-volume cylinder to allow amplification of muscle force or displacement. For a constant volume cylinder of radius r and length L , its volume is $V = \pi r^2 L$. A small decrease in radius produces a large increase in length proportional to $L \sim 1/r^2$ such that a 30% decrease in radius will result in the length roughly doubling.

The tongues of many terrestrial vertebrates — mammals and reptiles in particular — also commonly exploit this mechanical advantage. Tongues comprise a class of muscular hydrostats that are fascinating because of their widespread presence and exceptional level of diverse specialization across species. Some tongues, such as those of lizards and snakes, are specialized towards extension and retraction actions that aid in chemosensing. In this case, evolution has swapped the longitudinal and transverse muscle groups with the longitudinal muscle centrally located, known as hyoglossal bundles, and circular muscles surrounding them

to squeeze and elongate the tongue (Figure 2D).

Tongues, however, complicate clean distinctions between a muscular hydrostat and a musculoskeletal structure. More so than cephalopod appendages and elephant trunks, which are driven mostly from within, tongues are controlled through a combination of extrinsic muscles, which attach to the skull and so are musculoskeletal in nature, and intrinsic muscles, which lie within the tongue and operate according to muscular hydrostat principles. Further, a tongue's presence in a mouth means it will often use the surrounding skull to constrain its action and increase mechanical advantage, such as in swallowing, where the tip of the tongue leverages against the roof of the mouth to provide support. For an awkward time, try swallowing without placing your tongue on the roof of your mouth!

Controlling a muscular hydrostat

As hopefully clear by now, muscular hydrostats exhibit a high degree of biomechanical complexity that makes them difficult to control. Again consider your arm. Your biceps and triceps work together to set the angle of your elbow, which can be described by a single angle. This is known as a single kinematic degree of freedom. Your arm as a whole (ignoring your hand) has seven degrees of freedom that roughly relate to the joint angles of your shoulder (three degrees of freedom), elbow (one degree of freedom), and wrist (three degrees of freedom). Unless something has gone horribly wrong, these three joints are connected by rigid, straight links (your bones). Together, this means that controlling your arm in space consists of coordinating muscle activations to control these angles, known as a seven kinematic degrees of freedom control problem.

Muscular hydrostats are not a series of joints and links like your arm but are rather continuously deformable. Mathematically, this means having an infinite number of degrees of freedom (compared to your arm's seven), which dramatically complicates the neuromuscular control problem. Now, of course, no muscular hydrostat *actually* has an infinite number of degrees of freedom, though an elephant does have ~90,000 individual

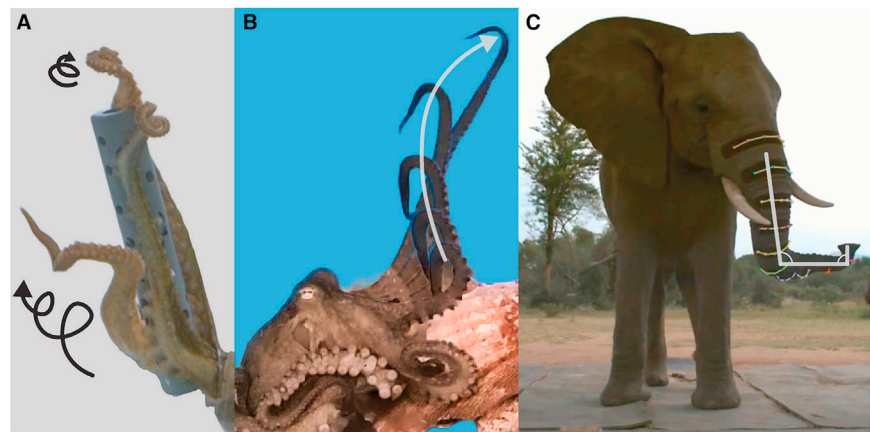


Figure 3. Muscular hydrostats employ a number of strategies to simplify their control.

(A) The muscular architecture of an octopus arm (*O. rubescens*) allows formation of complex shapes such as a helical corkscrew through co-contraction of only two muscle groups (longitudinal and oblique). (B) Octopuses (*O. bimaculoides*) reach for objects by forming a localized bend at the base of their arm and then propelling the bend along the arm. Adapted from Wang *et al.* (2025) with permission from Springer Nature. (C) African elephants (*Loxodonta africana*) form pseudo-joints of localized bends connected by straight, stiffened segments when moving the tip of their trunks from point to point. Reused from Dagenais *et al.* (2021).

muscle fascicles in its trunk. However, tongues, trunks, and tentacles exhibit a high degree of dexterous, fine-grained control all along their length such that they appear to possess an *effectively* infinite number of degrees of freedom. Solving this control task is a challenge, as any toddler learning to speak knows well.

Unsurprisingly, animals generally do not appear to solve this full infinite degree of freedom problem, but rather employ behavioral strategies that reduce the control complexity. One common strategy is to exploit the compliance inherent in a muscular hydrostat's soft, boneless structure to interact with, wrap around, and otherwise conform to objects in their environment. This allows a muscular hydrostat to forgo precise positional control, instead allowing the object to supply the shape by conforming to it.

I will focus here on the control of octopus arms and elephant trunks, as they represent perhaps the hardest version of the muscular hydrostat control problem because of their immense finesse and dexterity. Animals that primarily extend and retract a muscular hydrostat, such as snakes and their tongues, have much simpler control needs. It is important to first recognize how the control of muscular hydrostats is heavily mediated, and in many ways simplified, by the architectural

arrangement of their different muscle groups. This muscle architecture forms a mechanical program that generates functionally relevant shapes through simple combinations of muscle groups. For example, models have shown an octopus arm can generate a helical corkscrew shape by simply activating one of its four longitudinal muscle groups and its oblique muscles (Figure 3A). This combined activation of just two muscle groups also naturally aligns the arm's suckers to be oriented either inside (useful for grasping) or outside (useful for sensing) of the generated helix depending on which longitudinal muscle group is activated.

This mechanical program is critical to enabling another key control strategy, which is the propagation of waves of curvature along its length and requires muscle architectures that facilitate such continuous transport. Notably, this strategy has been observed in both octopus and elephants. When an octopus reaches for something far from its body, it will often form a bend at the base of its arm and then propel this bend along the arm to reach toward the object (Figure 3B). This propagating wave of bending has been correlated with electromyography measurements of a wave of muscle activation that similarly travels along the arm. Elephant trunks also utilize traveling waves of curvature during object manipulation and retrieval, suggesting a similar

use of waves of muscle activation and providing a fascinating example of seeming convergence in control strategy between these creatures.

A second strategy employed by both octopuses and elephants is the use of what has been termed pseudo-joints. Here, the animal generates localized bending at specific points along its length connected by straight, stiffened segments. This combination of localized joints and rigid links causes the arm or trunk to temporarily act like a rigid skeletal system, similar to your arm. Elephants employ this strategy when moving their trunk from point to point (Figure 3C), while octopuses use it to transport food to their mouths. Both actions require accurate control of the muscular hydrostat's tip, suggesting that when positional accuracy is important, simplifying the arm or trunk to mimic a vertebrate skeleton structure may be necessary to reduce control complexity enough to achieve precise motions.

Open questions

How sensing integrates into this control picture to help facilitate feedback control, and sensorimotor control in general, is an area I have mostly avoided here due to its complexity, though there is a long tradition of its study in neuroscience reaching back to J.Z. Young. Indeed, while some of the kinematic strategies used are well identified, the neuromuscular coupling that underlies muscular hydrostat control continues to be worked out. Despite the similarity in some of the behavioral strategies used by elephants and octopuses, their neural organization is quite dissimilar. An elephant trunk is controlled by a centralized nervous system, with more facial motor neurons than any other land mammal and a large share of them dedicated to the trunk. The control of octopus arms is instead highly decentralized, with approximately two-thirds of the octopus' neural tissue located in its arms and capable of executing local, peripheral motor programs not directly controlled by the central brain, making their behavioral convergence all the more striking.

I have focused here on muscular hydrostats that can be broadly viewed as slender cylinders. However, it is worth noting that there are other muscular hydrostats that do not exhibit this form factor. Examples

are the snouts of manatees and suckers of cephalopods. Arranged along a muscular hydrostat tentacle or arm, suckers are themselves muscular hydrostats, utilizing the same mechanical principles but with muscular architectures fine-tuned to produce suction. A final striking example is the buccal mass of many molluscs, in particular that of sea slugs such as *Aplysia californica*, which dynamically reconfigures itself through coordinated motor activation to achieve multifunctional behaviors using the same set of muscles.

The examples given in this Primer have hopefully illustrated the biomechanical diversity of muscular hydrostats. While related to classical water-filled hydrostats, their soft solid tissue structure provides them unique and sophisticated biomechanics that enable their exceptional dexterity. While the basic principles of muscular hydrostats are understood, in all but the simplest cases questions regarding particular biomechanical details and especially how animals establish such excellent neuromuscular control over them remain under active investigation. Beyond biology, these investigations have also been highly relevant to robotics: reproducing the control capabilities of an octopus arm or elephant trunk robot has been a long-standing goal of robotics — à la Doc Ock from Spider-Man. However, biology has solved and integrated numerous challenges that current engineering practice has yet to, making muscular hydrostats excellent examples of how biology can inform and inspire engineering practice. Despite immense progress, only a handful of muscular hydrostats have been studied in detail, and even then, often in only a few species. Studies of octopus arms have focused on only a small fraction of the roughly 300 recognized species and the immense diversity of animal tongues remains to be fully examined. As a result, there continues to be much to be discovered about muscular hydrostats, making these unique structures an exciting area of continued study for those of us captured by their mechanical beauty.

DECLARATION OF INTERESTS

The author declares no competing interests.

FURTHER READING

- Bennice, C.O., Buresch, K.C., Grossman, J.H., Morano, T.D., and Hanlon, R.T. (2025). Octopus arm flexibility facilitates complex behaviors in diverse natural environments. *Sci. Rep.* 15, 31875.
- Dagenais, P., Hensman, S., Haechler, V., and Milinkovitch, M.C. (2021). Elephants evolved strategies reducing the biomechanical complexity of their trunk. *Curr. Biol.* 31, 4727–4737.
- Flash, T., and Zullo, L. (2023). Biomechanics, motor control and dynamic models of the soft limbs of the octopus and other cephalopods. *J. Exp. Biol.* 226, jeb245295.
- Gutfreund, Y., Flash, T., Fiorito, G., and Hochner, B. (1998). Patterns of arm muscle activation involved in octopus reaching movements. *J. Neurosci.* 18, 5976–5987.
- Kier, W.M., and Smith, K.K. (1985). Tongues, tentacles and trunks: The biomechanics of movement in muscular-hydrostats. *Zool. J. Linn. Soc.* 83, 307–324.
- Kier, W.M., and van Leeuwen, J.L. (1997). A kinematic analysis of tentacle extension in the squid *Loligo pealei*. *J. Exp. Biol.* 200, 41–53.
- Kier, W.M., and Smith, A.M. (2002). The structure and adhesive mechanism of octopus suckers. *Integr. Comp. Biol.* 42, 1146–1153.
- Kier, W.M. (2012). The diversity of hydrostatic skeletons. *J. Exp. Biol.* 215, 1247–1257.
- Longren, L.L., Eigen, L., Shubitidze, A., Lieschnege, O., Baum, D., Nyakatura, J.A., Hildebrandt, T., and Brecht, M. (2023). Dense reconstruction of elephant trunk musculature. *Curr. Biol.* 33, 4713–4720.
- Marshall, C.D., Clark, L.A., and Reep, R.L. (1998). The muscular hydrostat of the Florida manatee (*Trichechus manatus latirostris*): A functional morphological model of perioral bristle use. *Mar. Mammal Sci.* 14, 290–303.
- Schulz, A.K., Boyle, M., Boyle, C., Sordilla, S., Rincon, C., Hooper, S., Aubuchon, C., Reidenberg, J.S., Higgins, C., and Hu, D.L. (2022). Skin wrinkles and folds enable asymmetric stretch in the elephant trunk. *Proc. Natl. Acad. Sci. USA* 119, e2122563119.
- Schwenk, K. (2025). Lizard and snake tongues as muscular hydrostats: Morphology, function, and diversity. *Integr. Comp. Biol.* 65, 1366–1380.
- Sekhavi, Y., Sellers, K.C., and Ross, C.F. (2026). Biomechanics and evolution of the primate tongue. *Evol. Anthropol.* 35, e70026.
- Sleboda, D.A., and Roberts, T.J. (2020). Internal fluid pressure influences muscle contractile force. *Proc. Natl. Acad. Sci. USA* 117, 1772–1778.
- Smith, K.K., and Mackay, K.A. (1990). The morphology of the intrinsic tongue musculature in snakes (Reptilia, Ophidia): Functional and phylogenetic implications. *J. Morphol.* 205, 307–324.
- Sumbre, G., Gutfreund, Y., Fiorito, G., Flash, T., and Hochner, B. (2001). Control of octopus arm extension by a peripheral motor program. *Science* 293, 1845–1848.
- Sumbre, G., Fiorito, G., Flash, T., and Hochner, B. (2006). Octopuses use a human-like strategy to control precise point-to-point arm movements. *Curr. Biol.* 16, 767–772.
- Tekinap, A., Naughton, N., Kim, S.H., Halder, U., Gillette, R., Mehta, P.G., Kier, W., and Gazzola, M. (2024). Topology, dynamics, and control of a muscle-architected soft arm. *Proc. Natl. Acad. Sci. USA* 121, e2318769121.
- Wang, T., Halder, U., Gribkova, E., Gillette, R., Gazzola, M., and Mehta, P.G. (2025). Neural models and algorithms for sensorimotor control of an octopus arm. *Biol. Cybern.* 119, 25.
- Ye, H., Morton, D.W., and Chiel, H.J. (2006). Neuromechanics of multifunctionality during rejection in *Aplysia californica*. *J. Neurosci.* 26, 10743–10755.

Department of Mechanical Engineering, Virginia Tech, Blacksburg, VA 24061, USA.
E-mail: nnaughton@vt.edu